

# The Planetary Power Loop

*An Orbital Solar-Power Constellation Beaming Continuous Energy to a Surface Colony*

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Classification: Speculative engineering, constrained to known physics

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## Abstract

A colony's most fragile organ is its power supply. On the Moon the sun sets for fourteen days at a time; on Mars, planet-wide dust storms can dim the sky for weeks, and it was such a storm that finally killed the Opportunity rover. Surface solar power, the obvious choice, is therefore also an intermittent and vulnerable one. The Planetary Power Loop removes the surface — and its weather and its nights — from the equation. The PL-1 'Helios Ring' is a constellation of orbital collectors that gather sunlight above the atmosphere around the clock, convert it to a microwave beam, and transmit it down to simple rectenna farms on the colony's surface, which turn it back into electricity. The beam shines through storm and darkness alike; the ground hardware is little more than wire mesh. We describe the collection, transmission, and rectification chain, the beam-safety design, the orbital architecture that maintains continuous coverage, and a staged path from a small demonstrator to a base-scale grid. Space-based solar power has crossed from theory to orbital demonstration; on another world, where the alternatives are worse, it may arrive first.

## 1. Introduction: The Sun Always Shines Somewhere Above

Electricity is the quiet precondition of everything a colony does — breathing, heating, growing food, making fuel, staying in touch with home — and losing it is not an inconvenience but a countdown. This is why the intermittency of surface solar power is so dangerous off Earth. The lunar night is a fourteen-day power outage. A Martian dust storm can blot the sun for weeks across a whole hemisphere, and one such storm ended the Opportunity mission after fifteen years. A settlement that depends on panels in the dirt is a settlement one long night or one bad storm away from crisis.

Yet just above the atmosphere, and always somewhere along a well-chosen orbit, the sun never stops shining. The idea of collecting that unbroken sunlight in space and sending the energy to where it is needed is not new — Peter Glaser proposed solar power satellites for Earth in 1968 (Glaser, 1968), and the concept has been studied continuously since (Mankins, 1997). What has changed recently is that the key step — transmitting usable power wirelessly from orbit to a receiver — has actually been demonstrated in space (Jaffe and McSpadden, 2013; and subsequent orbital tests). The Power Loop applies this to the one setting where its economics are most favourable: a colony that has no fossil reserves, no existing grid, and no tolerance for the dark.

## 2. Concept Description

The PL-1 'Helios Ring' is a constellation rather than a single station, and that is the heart of the design: several large collector satellites are spaced around an orbit so arranged that at least one always has both the sun in view and the colony below it, handing the beam from one to the next as they pass, the way relay runners pass a baton. Each collector is a broad, lightweight structure of thin-film photovoltaics or

concentrating reflectors that gathers raw sunlight and feeds a transmitter, which converts the electricity into a tightly-aimed microwave beam directed at a fixed point on the surface.

On the ground waits the humblest, most robust part of the whole system: the rectenna, a 'rectifying antenna' — in essence a large field of wire mesh and simple diodes that catches the microwave beam and converts it directly back into direct-current electricity. A rectenna has no moving parts, tolerates dust and cold, and can be built from local or easily-shipped materials over a wide, forgiving area. The colony's power plant, in this scheme, is a patch of mesh on the ground and a ring of collectors in the sky, joined by an invisible beam that neither night nor storm can interrupt.

### **3. The Chain from Sunlight to Socket**

The energy makes four conversions. In orbit, photovoltaics or a solar-thermal cycle turn sunlight into electricity; a transmitter then converts that electricity into microwaves, typically in a band chosen to pass cleanly through any atmosphere with minimal loss. The beam crosses the vacuum or thin air to the surface, where the rectenna converts it back to electricity for the grid. Each conversion costs efficiency, and the end-to-end figure is modest — but the input is free, continuous, and immune to weather, which for a colony matters more than raw efficiency (Mankins, 1997; Jaffe and McSpadden, 2013).

The beam's defining property is that it is deliberately gentle. Space-solar designs use a large transmitting aperture and a large receiving area precisely so that the power is spread thin: the peak intensity at the rectenna is engineered to be modest — comparable to sunlight, not a death ray — so that the beam is safe to stand in and harmless to anything that wanders across it (Jaffe and McSpadden, 2013). Safety is enforced by design and by control: the beam is kept locked onto the rectenna by a pilot signal broadcast from the ground, and if that signal is ever lost, the beam defocuses and shuts down. The system cannot aim where the receiver is not.

### **4. The Machine Itself**

The orbital segment's challenge is scale and structure. Collectors must be large and light, deployed or assembled in orbit from compact launches, and held rigid and accurately pointed while gathering sunlight and aiming a beam at a moving planet's surface. The transmitter is the sophisticated core: a phased array that forms and steers the microwave beam electronically, keeping it precisely on the ground station without physically slewing the whole structure. Thermal management, station-keeping, and the sheer area of photovoltaic material are the pacing engineering problems, all of them familiar in kind from large communications satellites, merely larger in degree.

The ground segment is deliberately the opposite: cheap, dumb, and durable. A rectenna is mostly open structure and mesh, spread over a wide area at low cost, with no optics to keep clean and no mechanisms to seize in the cold. This asymmetry is the concept's great practical virtue on a hostile world — the complexity lives in orbit, where there is no dust and no night, while the surface installation that the colonists must maintain is as close to maintenance-free as a power plant can be. Redundancy is built into the constellation: lose one collector and the others keep the beam alive, so the grid degrades gracefully rather than failing all at once.

### **5. Operations Concept**

Phase one is a single small collector in orbit and a modest rectenna on the surface, demonstrating the full chain — collect, beam, rectify — at kilowatt scale and proving beam safety and pointing in the real environment. Phase two expands to a partial constellation delivering continuous power to a growing base, sized to run life support, greenhouses, and in-situ fuel production through night and storm. Phase three completes the ring: enough collectors for uninterrupted, redundant, base-scale power, and — on the Moon especially — the option to beam energy directly into the permanently shadowed craters where the ice is mined but the sun never reaches, marrying this concept to the Cryo-Volatile Harvester elsewhere in this catalogue.

| Parameter       | Phase 1 (Demo)              | Phase 3 (Base grid)         |
|-----------------|-----------------------------|-----------------------------|
| Orbital segment | 1 small collector           | constellation (ring)        |
| Power delivered | kilowatt-class              | megawatt-class              |
| Coverage        | intermittent pass           | continuous (relay hand-off) |
| Ground segment  | small rectenna              | large rectenna field(s)     |
| Beam            | low-intensity, pilot-locked | same, scaled                |
| Key virtue      | prove safety + pointing     | immune to night + storm     |

*Table 1. Representative parameters for the staged build-out.*

## 6. Honest Difficulties

Three problems dominate. The first is mass and assembly: collectors large enough to matter require launching and building substantial structures in orbit, historically the reason space solar power has struggled to close its business case on Earth (Mankins, 1997). The second is end-to-end efficiency: four conversions each shed energy, so a great deal of collected sunlight never reaches the socket — tolerable when the input is free and the alternative is no power at all, but a real design pressure. The third is precision and safety at scale: keeping a beam locked on a rectenna across the geometry of a spinning world, with fail-safe defocus if pointing is lost, demands exacting control (Jaffe and McSpadden, 2013). None of these is a barrier of physics — wireless power transfer from orbit has been demonstrated — but each is a serious engineering and economic challenge, and the honest reason the concept may debut off Earth is simply that a colony has no cheaper option and cannot afford the dark.

## 7. Pathway to Reality

The concept has recently crossed a crucial threshold. Wireless power transmission is mature on the ground and has now been demonstrated from orbit: an experiment beamed measurable power from a satellite toward Earth, turning the central premise from analysis into flight heritage (building on the long-studied designs of Glaser, 1968; Mankins, 1997; Jaffe and McSpadden, 2013). Both national agencies and private ventures are actively developing space-based solar power for terrestrial use, and every advance there — lightweight collectors, efficient transmitters, in-orbit assembly — transfers directly to a planetary version. A kilowatt-scale demonstrator serving an early base is a credible mid-term project; a full ring follows the growth of the settlement it powers. The colony that first tires of the lunar night may be the one that finally makes space solar power pay.

## 8. Conclusion

On Earth, space-based solar power competes against cheap grids and rarely wins. On another world, it competes against darkness, and there it is peerless. The Planetary Power Loop is a bet that the decisive infrastructure of an off-world settlement is not a bigger battery or a nuclear pile alone but a ring of collectors in the sky, pouring uninterrupted sunlight down an invisible beam onto a humble field of mesh — power that ignores the storm, outlasts the night, and asks almost nothing of the people it keeps alive. When a colony no longer fears the dark, it has stopped merely surviving and begun to live.

## References

*Harvard referencing style (Cite Them Right).*

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